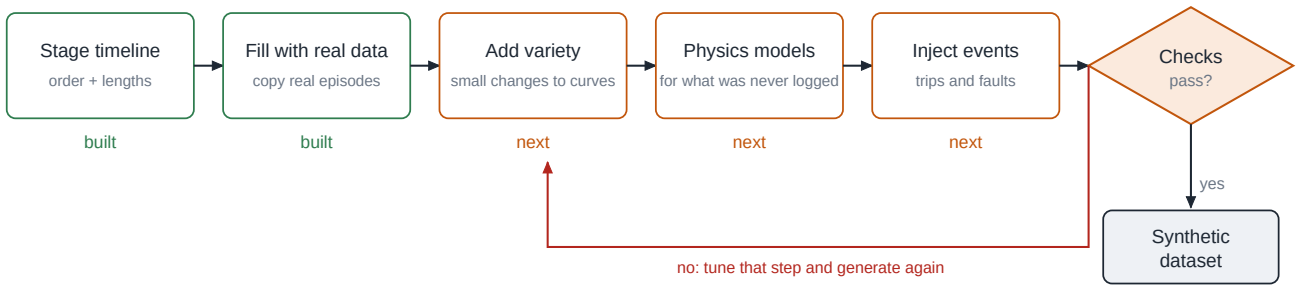
SOFC Fake Data Methods

> For this plant the best fake data is built in layers. First a stage timeline, then real recorded episodes copied into it, then small changes for variety, then simple physics for what was never recorded, then trips and faults added on purpose. The first two layers are already built.

Why it matters: you have 39 days of logs and only 2 to 26 real examples of each stage. Methods that learn from thousands of runs (such as GANs, a type of AI that learns to draw new curves) cannot work with that. Methods that reuse and reshape the real runs can.

The five layers



Green is built today (the stage pattern generator and the pattern filler). Orange is the work ahead. Anything that fails the checks goes back to be tuned.

All the methods at a glance

Method	What it does, in plain words	Fit for this plant
Stage timeline + real episodes	Draws a random but allowed order of stages, then fills each block with a real recorded episode	**Good fit** built: pattern generator and filler
Small changes to curves	Stretches a real curve in time, shifts it, or adds a little sensor noise	**Good fit** cheap variety, keeps the run's order
Physics models	Simple plant laws: heat-up speed, cooling curve, voltage falls as current rises	**Good fit** the only way to make things never recorded
Event injection	Adds a trip or fault on purpose, copying how the 8 real trips unfolded	**Good fit** rare events on demand
Reshuffle wiggles	Keeps a real trend but swaps its small wiggles with wiggles cut from other runs	**Limited** for long calm stages like Hot standby
Statistical time models	A formula learns how sensors move together minute by minute	**Limited** needs long calm stretches
SMOTE	Mixes two rare rows to make a third, for a training table	**Limited** only for a classifier, see the SMOTE pages
Deep generators (GAN, VAE, diffusion)	An AI model learns to draw brand-new curves	**Poor fit** needs hundreds of runs per stage
Full plant simulator	An engineering model of the whole plant (digital twin)	**Limited** most realistic, but weeks of work and design data

- **The method menu**: how much data each method needs, and how small changes work
- **Which method for which sensor**: the stack drawing, sensor by sensor
- **Build order**: what is done, what comes next, decisions for you
- **SMOTE deep-dive**: the earlier SMOTE guide, kept as 4 pages

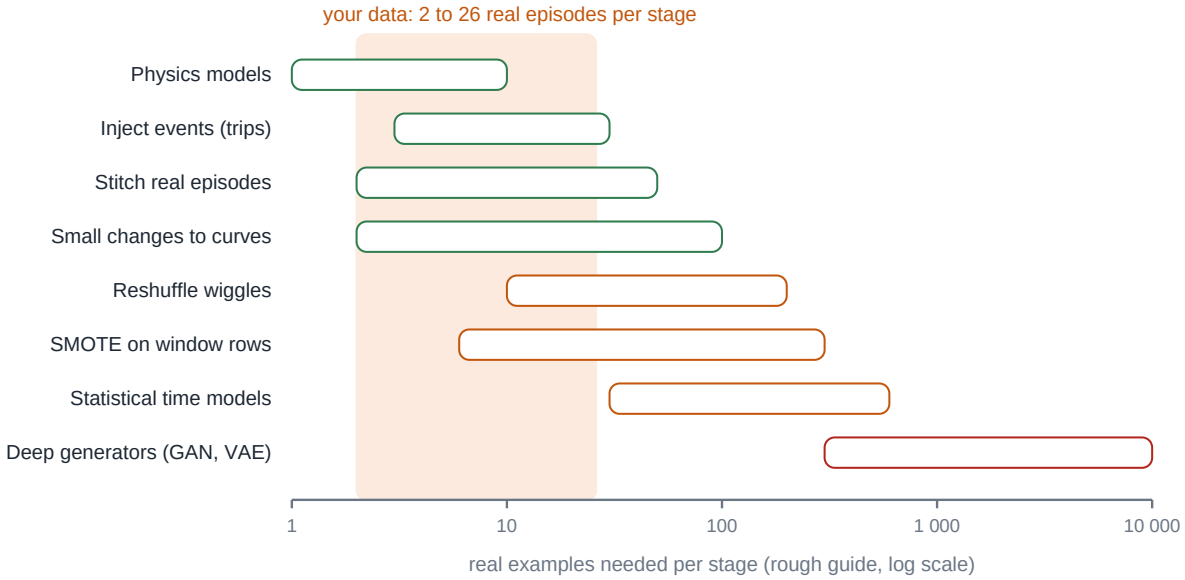
IF YOU REMEMBER ONE THING

With so few real runs, reuse and reshape the real data, and use simple physics only where nothing was recorded.

The Method Menu

Every method needs a certain number of real examples to work. Your data sits at 2 to 26 examples per stage, so the methods on the left of the chart fit and the AI generators on the right do not.

How much real data each method needs

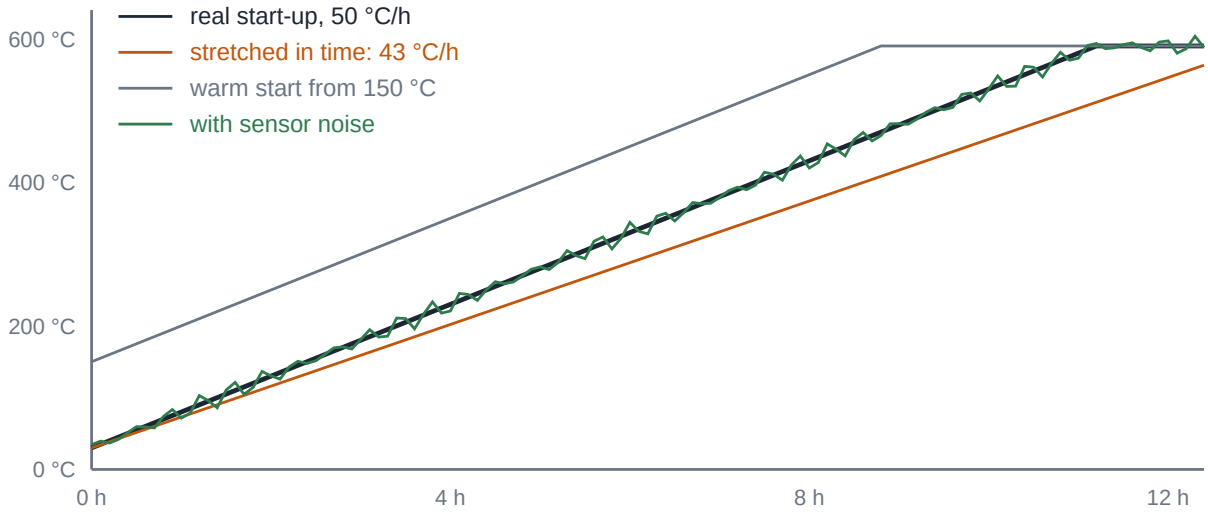


Each bar shows roughly how many real examples per stage a method needs. The orange band is your data. Green bars fit, orange partly fit, red does not. These are rules of thumb, not exact limits.

Your real examples per stage

Stage	Real episodes	What that allows
Hot standby	26	copy, small changes, reshuffle wiggles
Shut-down	19 (6 full cool-downs)	copy pieces; cooling law for new speeds
Thermal adjustment	18	copy and stretch
Start-up	14	copy and stretch; heat-up ramp rules
Part load	10 runs, 8 end in a trip	copy, stretch; voltage-current curve above 16 A
Idle	2	copy only
Rated power	0	physics model only, or leave out

What 'small changes' look like



Stack temperature during heat-up: one real curve and three small changes

Press Play: one real start-up (dark) becomes three new ones. Each stays inside the plant's limits, such as the heater ramp of about 60 K/h.

The pattern filler already does two of these: it stretches Start-up, Thermal adjustment and Part load blocks in time, and it enters a Start-up or cool-down part-way when the stack is already warm. Noise and gentle scaling are the next easy step.

Physics models in one line each

Law	What the data says	Used for
Heat-up	heater outlet ramps about 60 K/h; the stack follows at about 50 °C/h	Start-up of any length
Cool-down	temperature falls along a curve with time constant about 21 h at 6000 ln/min air, 5 to 9 h at 8000 to 10000	Shut-down at new air flows
Voltage vs current	each string: about 480 to 490 V with no load, minus about 4 to 6 Ω times the current	Part load above 16 A and rated power, if wanted
Natural cooling	air off: time constant about 89 h (200 to 50 °C in about 6 days)	long plant-off stretches

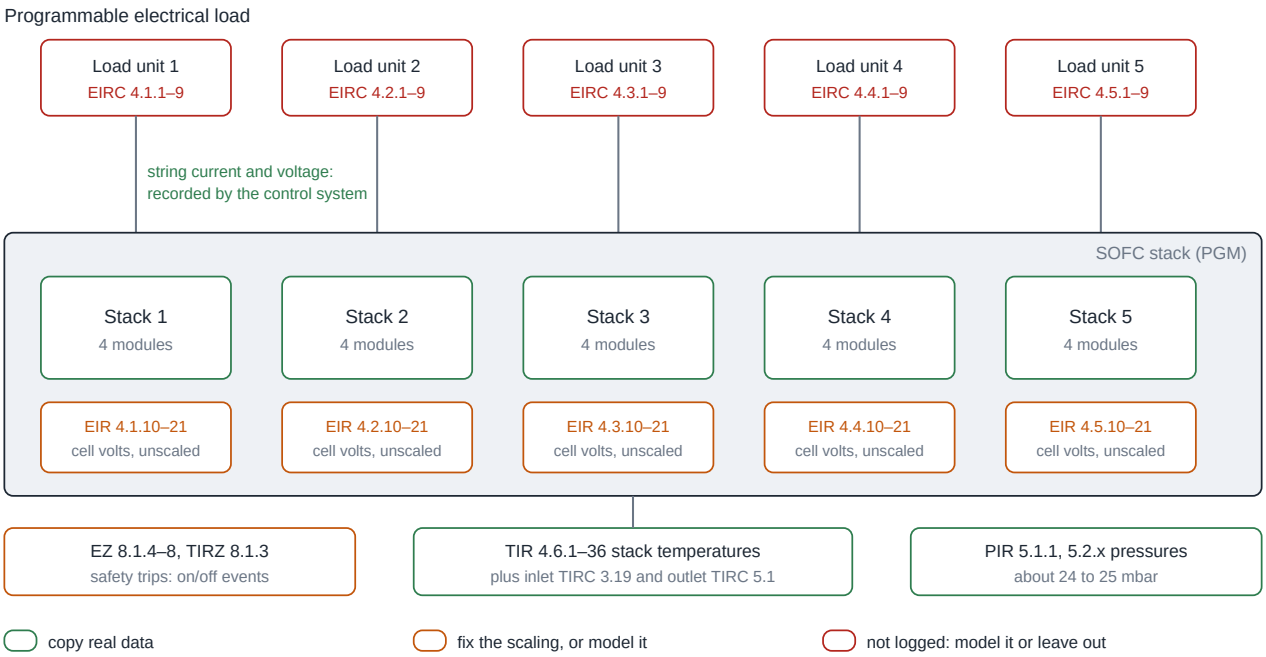
IF YOU REMEMBER ONE THING

Choose methods that fit 2 to 26 examples: copy, stretch, add noise, apply physics. Leave AI generators until there are hundreds of runs.

Which Method, Which Sensor

Most sensors in the stack drawing were recorded well and can be copied from real episodes. Two groups cannot: the load-unit signals were never logged, and the cell voltages look unscaled. Those need a fix at the source or a physics model.

The stack area, coloured by method



Redrawn from your plant drawing. Press Play to go through it group by group.

Sensor by sensor

Sensor group	Tags in the drawing	State of the data	Method
Stack temperatures	TIR 4.6.1 to 4.6.36, inlet TIRC 3.19.x, outlet TIRC 5.1.x	logged every second, complete	**Copy real** plus small changes
Pressures	PIR 5.1.1, PIR 5.2.x	logged, about 24 to 25 mbar	**Copy real** light noise only
String current and power	control-system tags behind the load lines	logged; part load up to 16.2 A and 33.6 kW	**Copy real** physics above 16 A
Cell / module voltages	EIR 4.1.10 to 4.5.21	values near 0.005 with -1 fills, not real volts	**Fix or model** ask the plant team for the scaling
Load-unit signals	EIRC 4.1.1 to 4.5.9 (45 tags)	not in the logs at all	**Not logged** work out from string current, or leave out
Safety trips	EZ 8.1.4 to 8.1.8, TIRZ 8.1.3	on/off events, 8 trips in part load	**Inject by rule** exhaust 200 °C or more means a trip

Copy whole moments, never single sensors. The filler takes all 175 sensors from the same real minute, so the links between them (for example, outlet hotter than inlet under load) stay true. Mixing sensors from different runs breaks those links.

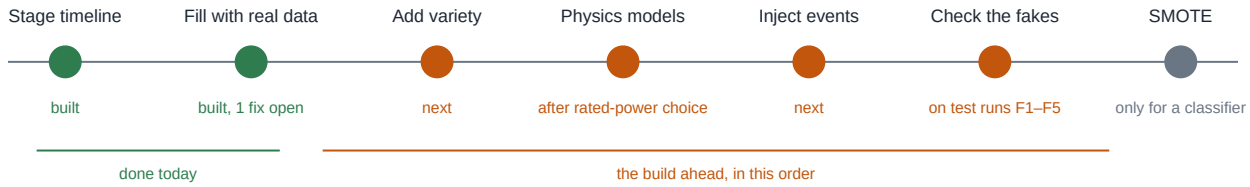
IF YOU REMEMBER ONE THING

Copy what was logged. For the load units and the cell voltages, fix the source or model them; do not invent them freely.

Build Order

Two layers are built. Next come variety, physics, events and checks, in that order. SMOTE comes last, and only if you want a model that sorts or warns.

The plan



Press Play to walk the milestones in order.

Each step

Step	What to do	Status
1. Stage timeline	random allowed order of stages with real lengths	**Built** pattern_generator.py
2. Fill with real data	copy real episodes into each block	**Built** pattern_filler.py; open: draw Start-up length from start temperature (18 % of blocks flagged)
3. Add variety	noise, gentle scaling, time stretch, reshuffled wiggles in Hot standby	**Next**
4. Physics models	heat-up ramp, cooling law, voltage-current curve per string	**Next** rated power waits for your choice
5. Inject events	trips copied from the 8 real ones; set how often	**Next**
6. Check the fakes	plant limits, look-alike test, usefulness on real test runs F1 to F5	**Next**
7. SMOTE (optional)	balance window rows for a stage or trip classifier	**Next** see the SMOTE pages

Decisions only you can make

Decision	Options	Suggested start
Rated power	leave out, or build it from design values	leave out until design values are confirmed
Cell voltages EIR 4.x.10 to 21	get the scaling from the plant team, or model them	ask the plant team first
How much variety	small noise only, or also time stretch and scaling	noise and stretch within the real min and max
Trips in fake data	none, as often as real (8 of 10 part-load runs), or a set rate	as often as real
What the fake data is for	testing software, training a model, or demos	decide before step 6; it sets the checks

IF YOU REMEMBER ONE THING

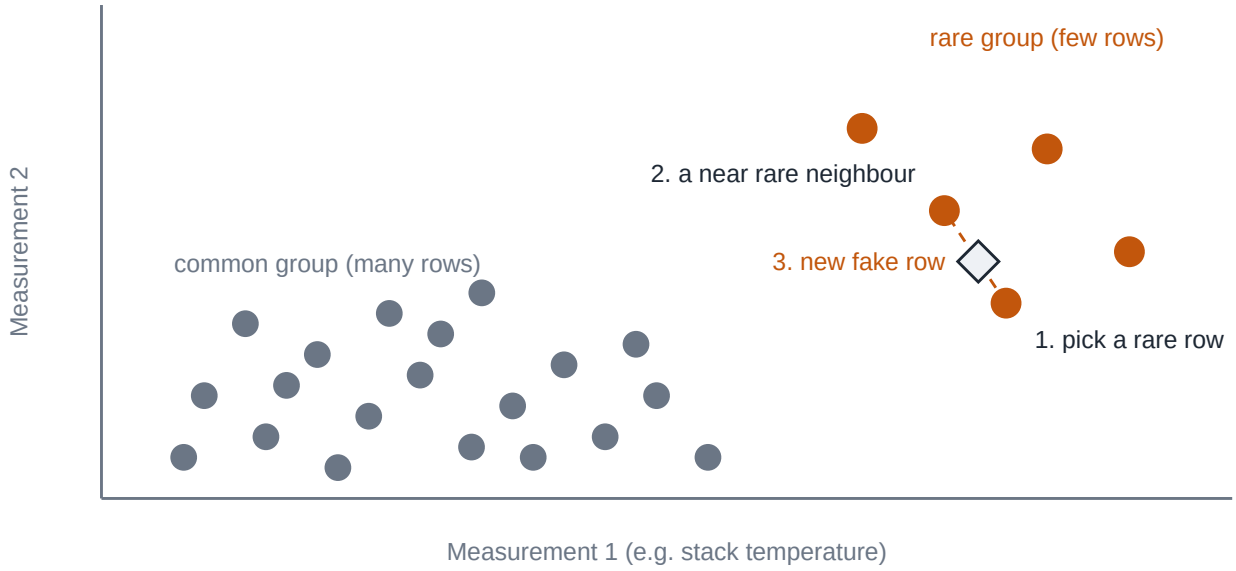
Finish the layers in order: variety, physics, events, checks. Each one adds realism without needing more real runs.

SMOTE Readiness Guide

To use SMOTE (a method that invents extra rows for a rare group by mixing two real rows of that group), you need five things: a clear job for the model, one labelled row per example, a test set locked away first, enough real examples of the rare group, and checks that throw out impossible fake rows.

Why it matters: fake rows only help if they teach the model something true. Without these ingredients they either teach the model things the plant never does, or they leak into the test and make the scores look better than they are.

How SMOTE makes one fake row



Every fake row lies on a straight line between two real rare rows. SMOTE never goes outside what it has already seen.

The short list

What you need	In plain words	Your project today
A job for the model	What question should the model answer? (e.g. which stage is the plant in, will it trip soon)	**Not yet**
Labels on every example	Each example says which group it belongs to	**Have it** stage labels from the approved stage rules
One row per example	Curves cut into time windows and summed up as numbers	**Not yet**
A locked test set	Some whole runs kept aside before any fake rows are made	**Have it** 5 run-based test folds (F1 to F5)
Enough rare examples	At least 6 real examples of the rare group, from different runs	**Partly** Part load 10 runs, full Shut-down only 5
A check for impossible rows	Plant limits that every fake row must respect	**Partly** limits known from the stage notebooks

The next three pages go through these one by one. They are the SMOTE deep-dive; the method overview starts on the home page.

- **What you need, step by step**: the full recipe and how many real examples you have
- **Why curves are tricky**: why SMOTE mixes up plant runs, and the other methods
- **Checking the fakes**: the three tests a fake row must pass, and your three decisions

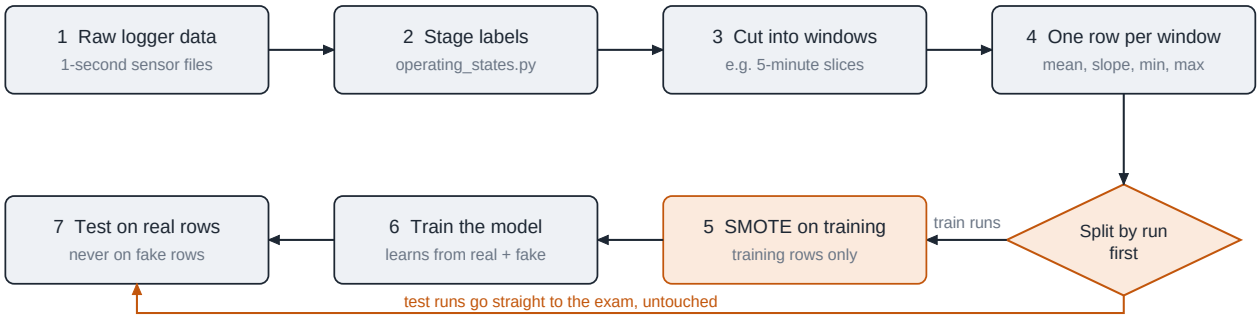
IF YOU REMEMBER ONE THING

SMOTE balances a training table by mixing rows you already have. It cannot create plant behaviour that was never recorded.

What SMOTE Needs

The order matters most: label the data, cut it into rows, set the test runs aside, and only then make fake rows, using the training part alone.

The recipe, in order



Seven steps. The orange diamond is the gate that matters: split by run before any fake rows are made.

Each ingredient, and where you stand

Need	Why	Your project today
1. A job for the model	SMOTE balances groups. You must know which groups, and that depends on the question.	**Not yet** to decide (see page 4)
2. Labels	SMOTE only mixes rows with the same label.	**Have it** 6 stages from the approved stage rules; trip events found in part load
3. One row per example	SMOTE works on a table. A 1-second curve has to be cut into windows (e.g. 5 minutes) and summed up: average, slope, lowest, highest.	**Not yet** not built yet
4. Clean numbers	SMOTE measures distance between rows. Gaps break it, and big-unit columns (temperature in °C) drown small ones (voltage).	**Partly** logger dropout minutes still to blank or fill; scaling still to add
5. A locked test set	Rows from one run look alike. If one run sits in both training and test, the test is too easy.	**Have it** folds F1 to F5 are split by run
6. Enough rare examples	The usual SMOTE setting looks at 5 neighbours, so it needs at least 6 real rare examples.	**Partly** see the table below
7. Plant limits	Used to throw out fake rows the plant could never produce.	**Partly** list on page 4
8. Software	The standard tool is the Python package imbalanced-learn.	**Not yet** not installed on this PC yet
9. A yardstick	Train the same model with and without fake rows and compare on the real test runs.	**Not yet** set up once the job is chosen

How many real examples do you have?

Stage	Real examples	Enough for SMOTE?
Hot standby	about 20 holds	**Enough**
Thermal adjustment	18 episodes	**Enough**
Start-up	15 episodes (8 cold, 7 warm)	**Enough** but cold and warm differ, so treat them apart
Part load	10 runs, 8 of them end in a trip	**Borderline** just enough; only 2 calm endings
Shut-down, full cool-down	5	**Borderline** too few for the usual setting; use 3 neighbours or fewer
Rated power	0	**None** SMOTE cannot make it; it only mixes what exists

Counting tip: count runs, not rows. Ten part-load runs cut into 5-minute windows give many rows, but SMOTE mixing two windows of the same run mostly copies that run.

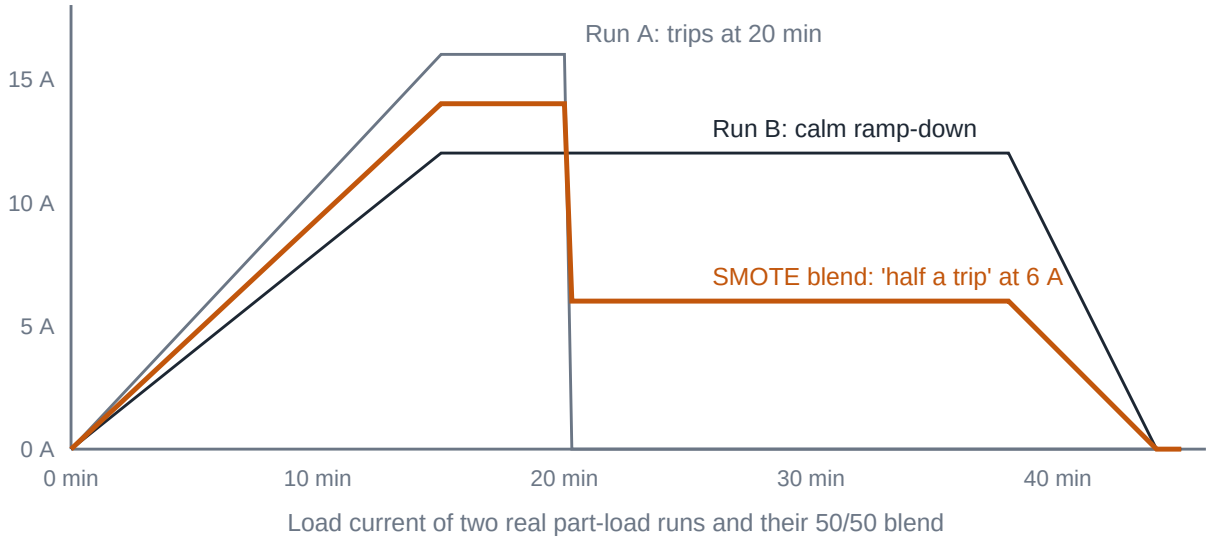
IF YOU REMEMBER ONE THING

Split the runs first. Make fake rows only inside the training part. Never test on fake rows.

Why Curves Are Tricky

SMOTE treats each row on its own, but a plant run is a story in time. Mixing two runs blends two stories into one that never happened.

What a blend looks like



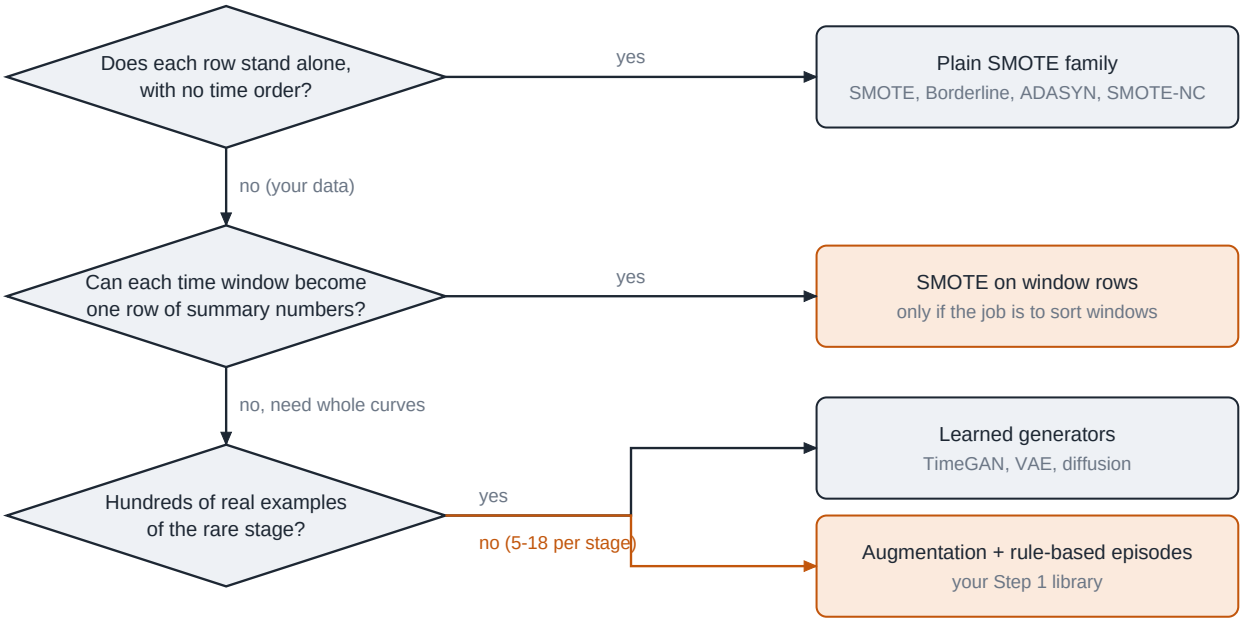
Two real part-load runs (grey and dark) and their blend (orange). The blend shows a 'half trip' that no plant can do.

The same happens with on/off signals. A heater that is on in one row and off in the other becomes 'half on' in the blend. And because SMOTE ignores time order, it can mix a minute from early heat-up with a minute from late heat-up.

The methods side by side

Method	What it does	Needs	Fit for your data
SMOTE	Mixes two rare rows	A table of numbers	**Limited** only on window summary rows
SMOTE-NC	Same, but copies on/off and text columns instead of averaging them	A mixed table	**Limited** use this one if on/off columns stay in
Borderline-SMOTE, ADASYN	Puts more fake rows near the border between groups	A table, low noise	**Limited** optional variant, try after plain SMOTE
Small changes to real curves	Adds a little noise, scales or stretches a real run in time	Real curves	**Good fit** simple, keeps the run's order
Learned generators (TimeGAN, VAE, diffusion)	A model learns to draw brand-new curves	Hundreds to thousands of runs	**Poor fit** far too few runs today
Rule-based episodes	Stitches real stage episodes in the allowed order and varies them within limits	Stage labels and rules	**Good fit** your current plan, Step 1 done

Which one to pick



Follow the questions from the top. Your data takes the orange paths.

IF YOU REMEMBER ONE THING

Use SMOTE on summary rows to balance a classifier. Use small changes and the rule-based plan to make whole new runs.

Checking the Fakes

A fake row is kept only if it passes three tests: it obeys the plant's limits, it looks like the real data, and it makes the model better on real test runs.

The three gates

Each fake row goes left to right. Failing any orange gate means drop it or rethink the method.

Gate 1: plant limits

Rule	Limit	Where it comes from
Heat-up speed	heater outlet ramp about 60 K/h; stack about 50 °C/h	Start-up notebook
Lowest cell voltage	not below 0.72 V while under load	operator limit seen in part load
Load current	up to 16.2 A, the highest seen	Part-load notebook (higher only if you decide on design values)
Exhaust temperature	at 200 °C or more, a trip must follow	trip analysis
On/off signals	exactly on or off, never in between	plant logic
Stage order	only the 18 allowed arrows of the stage map	stage map

Gate 2: does it look real?

Compare the spread of each sensor in real and fake rows. Then train a small 'spot the fake' model. If it can only guess right about half the time, the fakes look real.

Gate 3: does it help?

Train the same model twice on the same folds F1 to F5: once on real rows only, once with fake rows added. Score both on the real test runs. Keep the fake rows only if the score goes up. A moderate top-up (for example the rare group raised to half the size of the common one) is often safer than full balance.

Three decisions only you can make

Decision	Options	Suggested start
1. What job should the model do?	find the current stage; warn before a trip; estimate a sensor value	warn before a trip (trips are the rare, costly event)
2. How much to top up the rare group?	a little, half, or full balance	half, then test

Decision	Options	Suggested start
3. SMOTE, the rule-based plan, or both?	SMOTE for a classifier on window rows; rules for whole new runs	both, each for its own job

IF YOU REMEMBER ONE THING

Fake rows earn their place only by passing the plant limits, looking real, and improving results on real test runs.